



GOSFORD HIGH SCHOOL

**2016
TRIAL HSC EXAMINATION.**

MATHEMATICS

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Board approved calculators may be used
- Write using black pen
- A reference sheet is provided
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section I – 10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours 45 minutes for this section

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple choice answer sheet for Questions 1 -10.

1 What is 52.09684 correct to 4 significant figures?

- (A) 52.0968
- (B) 52.09
- (C) 52.10
- (D) 52.1

2 Which of the following is equal to $\frac{\sqrt{3}}{2\sqrt{3}+\sqrt{2}}$?

- (A) $\frac{6-\sqrt{6}}{4}$
- (B) $\frac{6-\sqrt{6}}{10}$
- (C) $\frac{6+\sqrt{6}}{10}$
- (D) $\frac{3-\sqrt{6}}{5}$

- 3 The quadratic equation $x^2 - 3x + 1 = 0$ has roots α and β . What is the value of $\alpha^2 + \beta^2$?

(A) 11
(B) 7
(C) 9
(D) -11

- 4 A geometric series has $T_1 = \log 3$ and $T_2 = \log 9$. If $T_3 = \log x$, what is the value of x ?

(A) 27
(B) 12
(C) 15
(D) 81

- 5 Let $a = e^x$. Which expression is equal to $\log_e(a^2)$?

(A) x^2
(B) e^{x^2}
(C) e^{2x}
(D) $2x$

6 What are the amplitude and period of the function $f(x) = 2 - \sin 2x$?

- (A) Amplitude 1, period π
- (B) Amplitude 1, period 2π
- (C) Amplitude 2, period π
- (D) Amplitude 2, period 2π

7 What is the value of

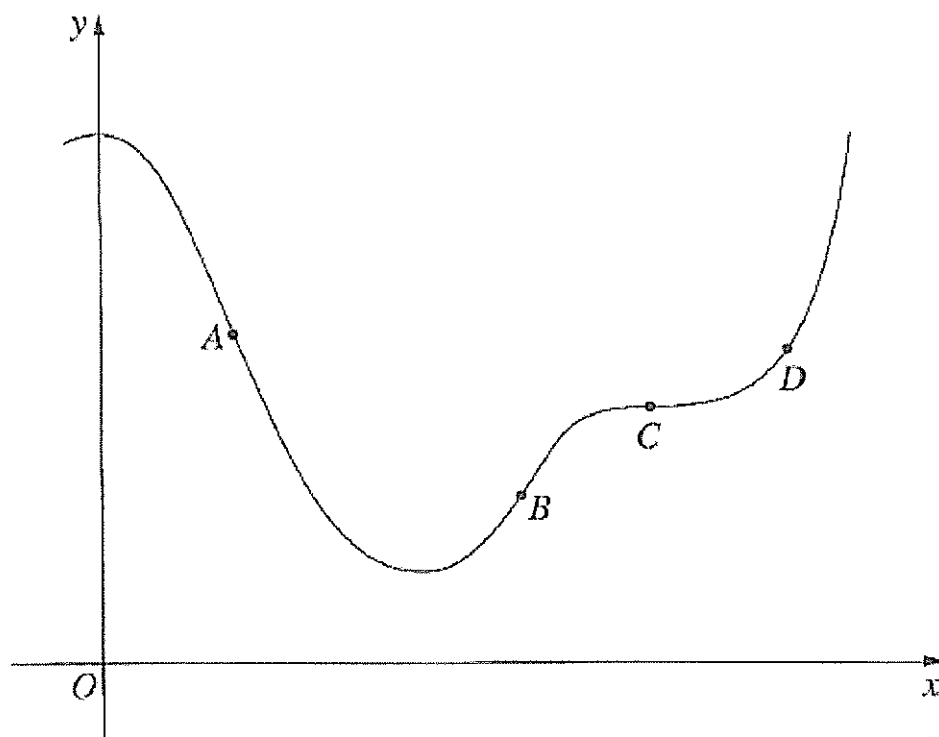
$$\sum_{k=1}^4 (-1)^k k^2$$

- (A) -30
- (B) -10
- (C) 10
- (D) 30

8 Which of the following trigonometric expressions is equivalent to $\tan\left(\frac{\pi}{2} - x\right)$?

- (A) $\tan x$
- (B) $\cot x$
- (C) $-\tan x$
- (D) $-\cot x$

- 9 The diagram shows the points A , B , C and D on the graph $y = f(x)$.



At which point is $f'(x) > 0$ and $f''(x) = 0$?

- (A) A
 - (B) B
 - (C) C
 - (D) D
- 10 A particle is moving along the x axis. The displacement of the particle at time t seconds is x metres.
- At a certain time, $\dot{x} = -3 \text{ ms}^{-1}$ and $\ddot{x} = -2 \text{ ms}^{-2}$.
- Which statement describes the motion of the particle at that time?
- (A) The particle is moving to the right with increasing speed.
 - (B) The particle is moving to the left with increasing speed.
 - (C) The particle is moving to the right with decreasing speed.
 - (D) The particle is moving to the left with decreasing speed.

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

Answer each question in a new writing booklet.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a new booklet.

(a) Factorise $40x^3 - 5$ 2

(b) Solve $|2x - 1| \leq 3$ 2

(c) Simplify

$$\frac{9^n \times 15^{2-2n}}{25^{1-n}}$$

2

(d) Differentiate with respect to x :

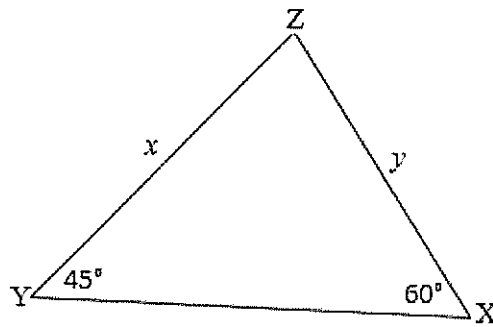
(i) $\ln(2x + 1)$ 1

(ii) $(e^{3x} + 2)^4$ 2

(iii) $\frac{\sin x}{x^3}$ 2

Question 11 continues on next page

(e)



In the diagram, XYZ is a triangle where $\angle ZYX = 45^\circ$ and $\angle ZXY = 60^\circ$.

Find the exact value of the ratio $\frac{x}{y}$.

2

(f) Solve $\log_2(3x - 4) = 5$

2

Question 12 starts on next page

Question 12 (15 marks) Start a new booklet.

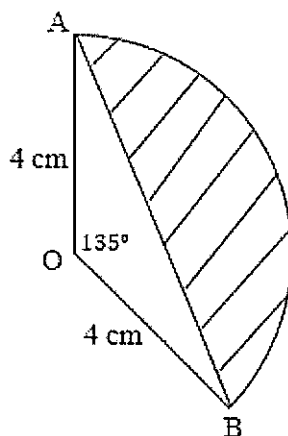
- (a) Find the coordinates of the focus of the parabola $x^2 = 12(y - 3)$ 2

- (b) Find $\int \frac{6x}{x^2-3} dx$ 2

- (c) Evaluate

$$\lim_{x \rightarrow 3} \frac{x^2 - 2x - 3}{x - 3}$$
2

- (d) The diagram shows a sector of a circle, centre O. AB is a chord of the circle.
If $\angle AOB = 135^\circ$, show that the area of the shaded region is $(6\pi - 4\sqrt{2})\text{cm}^2$. 2



- (e) If $f'(x) = 2x + 7$ and $y = f(x)$ passes through the point (1,4), find $f(x)$. 2

- (f) Find the equation of the tangent to the curve $y = 3x^2 + 2x + 1$ at the point (1,6). 3

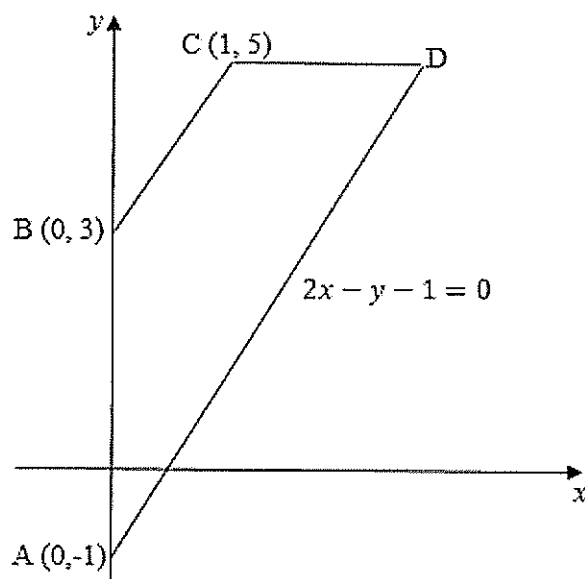
- (g) Solve the following for x :

$$e^{2x} + 3e^x - 10 = 0$$
2

Question 13 starts on next page

Question 13 (15 marks) Start a new booklet.

(a)



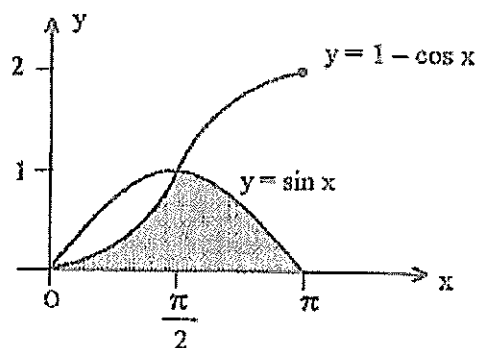
In the diagram, $ABCD$ is a quadrilateral. The equation of the line AD is

$$2x - y - 1 = 0.$$

- (i) Show that $ABCD$ is a trapezium by showing that BC is parallel to AD . 2
- (ii) The line CD is parallel to the x axis. Find the coordinates of D . 1
- (iii) Find the length of BC . 1
- (iv) Show that the perpendicular distance from B to AD is $\frac{4}{\sqrt{5}}$. 2
- (v) Hence, or otherwise, find the area of the trapezium $ABCD$. 2

Question 13 continues on next page

(b)



The diagram above shows the graphs of the functions $y = 1 - \cos x$ and $y = \sin x$, between $x = 0$ and $x = \pi$. The two graphs intersect at the point where $x = \frac{\pi}{2}$. Evaluate the area of the shaded region. 4

(c) If $y = \ln \left[\frac{1-x}{1+x} \right]$, show that $\frac{dy}{dx} = \frac{-2}{1-x^2}$.

Hence or otherwise, evaluate

$$\int_0^{\frac{1}{2}} \frac{dx}{1-x^2}$$

3

Question 14 starts on next page

Question 14 (15 marks) Start a new booklet.

- (a) (i) Show that $\cos\theta \tan\theta = \sin\theta$ 1
- (ii) Hence solve $8\sin\theta \cos\theta \tan\theta = \operatorname{cosec}\theta$ for $0 \leq \theta \leq 2\pi$ 2
- (b) A function is given by $f(x) = 3x^2 - x^3 + 9x - 2$.
- (i) Find the coordinates of any stationary points and determine their nature. 3
- (ii) Find the coordinates of any points of inflexion. 2
- (iii) Sketch the graph of $y = f(x)$ for $-2 \leq x \leq 5$. 2
- (iv) For what values of x over the given domain is the function concave up? 1
- (c) Find the volume of the solid formed when the area between the curve $y = \ln(2x)$ and the y axis is rotated about the y axis from $y = 1$ to $y = 6$. 4

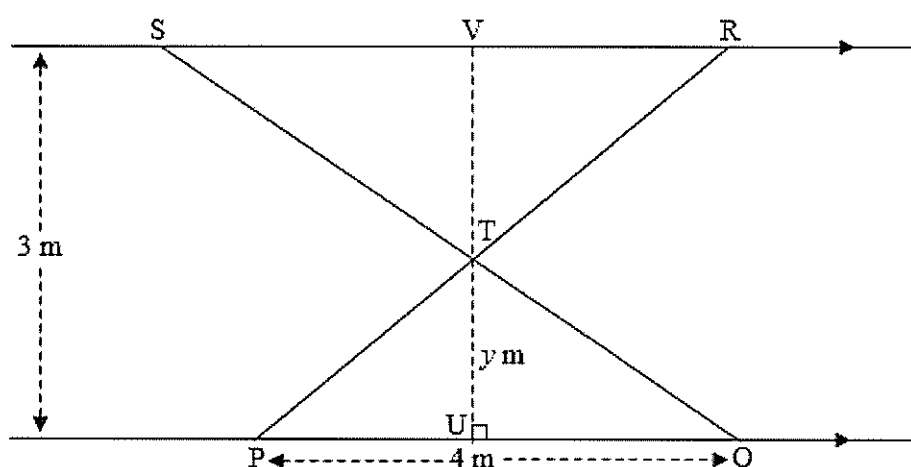
Question 15 starts on next page

Question 15 (15 marks) Start a new booklet.

(a) State the domain and range of the function $y = \sqrt{25 - x^2}$. 2

(b) Find the value(s) of k for which $x^2 - (k - 2)x + 3(k - 2) = 0$ has no real roots. 3

(c)



In the diagram, PQ and SR are parallel railings which are 3 metres apart. The points P and Q are fixed 4 metres apart on the lower railing. Two crossbars PR and QS intersect at T as shown in the diagram. The line through T perpendicular to PQ intersects PQ at U and SR at V . The length of UT is y metres.

(i) By using similar triangles, or otherwise, show that $\frac{SR}{PQ} = \frac{VT}{UT}$. 4

(ii) Show that $SR = \frac{12}{y} - 4$. 1

(iii) Hence, show that the total area, A , of $\triangle PTQ$ and $\triangle RTS$ is

$$A = 4y - 12 + \frac{18}{y} \quad \text{2}$$

(iv) Find the value of y that minimises A . 3

Question 16 starts on next page

Question 16 (15 marks) Start a new booklet.

- (a) (i) Use Simpson's rule with 3 function values to find an approximation to the area under the curve $y = \frac{1}{x}$ between $x = a$ and $x = 3a$, where a is positive. 2
- (ii) Using the result in part (i), show that $\ln 3 \div \frac{10}{9}$ 1
- (b) A particle is initially at rest at the origin. Its acceleration as a function of time, t , is given by $\ddot{x} = 4\sin 2t$.
- (i) Show that the velocity of the particle is given by $\dot{x} = 2 - 2\cos 2t$. 2
- (ii) Sketch the graph of the velocity $0 \leq t \leq 2\pi$ and determine the time at which the particle first comes to rest after $t = 0$. 3
- (iii) Find the distance travelled by the particle in the first $\frac{2\pi}{3}$ seconds. 2
- (c) On the 1st January 2000, Toby deposited \$15 000 into a bank account that paid interest at a fixed rate of 4% per annum compounded annually. He later decided to add \$5000 to his account on 1st January each year, starting on 1st January 2007.
- (i) Write an expression for the amount in the account on 1st January 2007 after the payment of interest and the first \$5000 deposit. 1
- (ii) How much was in Toby's account on 1st January 2016 after the payment of interest and the \$5000 deposit? 4

End of examination

Section I

2016 TRIAL HSC MATHEMATICS SOLUTIONS.

Q1 (C)

$$2. \frac{\sqrt{3}}{2\sqrt{3}+\sqrt{2}} \times \frac{2\sqrt{3}-\sqrt{2}}{2\sqrt{3}-\sqrt{2}}$$

$$= \frac{6-\sqrt{6}}{12-2}$$

$$= \frac{6-\sqrt{6}}{10} \quad (B)$$

$$3. \alpha + \beta = 3, \alpha\beta = 1$$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= 9 - 2$$

$$= 7 \quad (B)$$

$$4. T_1 = \log 3$$

$$T_2 = \log 3^2$$

$$= 2\log 3. \therefore r = 2$$

$$\log x = 4\log 3$$

$$= \log 3^4$$

$$\therefore x = 81 \quad (D)$$

$$5. \log_e(a^2) = 2\log_e a$$

$$= 2\log_e e^x$$

$$= 2x \quad (D)$$

6. (A)

$$7. \sum_{k=1}^4 (-1)^k k^2 = -1 + 4 - 9 + 16$$

$$= 10$$

(C)

8. (B)

9. (B)

10. (B)

Section II

Question 11

$$a) 40x^3 - 5 = 5(8x^3 - 1)$$

$$= 5(2x-1)(4x^2+2x+1)$$

$$b) |2x-1| \leq 3$$

$$2x-1 \leq 3 \text{ or } 2x-1 \geq -3$$

$$2x \leq 4$$

$$x \leq 2$$

$$2x \geq -2$$

$$x \geq -1$$

$$-1 \leq x \leq 2$$

$$c) \frac{9^n \times 15^{2-2n}}{25^{1-n}}$$

$$= \frac{3^{2n} \times 3^{2-2n} \times 5^{2-2n}}{5^{2-2n}}$$

$$= 3^2$$

$$= 9$$

$$d) i) \frac{d}{dx} \ln(2x+1) = \frac{2}{2x+1}$$

$$ii) \frac{d}{dx} (e^{3x} + 2)^4 = 4(e^{3x} + 2)^3 \cdot 3e^{3x}$$

$$= 12e^{3x} (e^{3x} + 2)^3$$

$$iii) \frac{d}{dx} \frac{\sin x}{x^3} = \frac{x^3 \cos x - 3x^2 \sin x}{x^6}$$

$$= \frac{x^2(x \cos x - 3 \sin x)}{x^6}$$

$$= \frac{x \cos x - 3 \sin x}{x^4}$$

$$e) \frac{x}{\sin 60^\circ} = \frac{y}{\sin 45^\circ}$$

$$\frac{x}{\sqrt{3}/2} = \frac{y}{1/\sqrt{2}}$$

$$\frac{2x}{\sqrt{3}} = \sqrt{2}y$$

$$\frac{x}{y} = \frac{\sqrt{6}}{2}$$

$$f) \log_2(3x-4) = 5$$

$$3x-4 = 2^5$$

$$3x-4 = 32$$

$$3x = 36$$

$$x = 12$$

Question 12

$$a) a = 3. \text{ Vertex } (0, 3)$$

$$\therefore \text{Focus } (0, 6)$$

$$b) \int \frac{6x}{x^2-3} dx$$

$$= 3 \int \frac{2x}{x^2-3} dx$$

$$= 3 \ln(x^2-3) + C$$

$$c) \lim_{x \rightarrow 3} \frac{x^2-2x-3}{x-3}$$

$$= \lim_{x \rightarrow 3} \frac{(x-3)(x+1)}{x-3}$$

$$= \lim_{x \rightarrow 3} (x+1)$$

$$= 4$$

d)

$$\begin{aligned}
 A &= \frac{1}{2} r^2 (\theta - \sin \theta) \\
 &= \frac{1}{2} \times 16 \left(\frac{3\pi}{4} - \sin \frac{3\pi}{4} \right) \\
 &= 8 \left(\frac{3\pi}{4} - \frac{1}{\sqrt{2}} \right) \\
 &= 6\pi - \frac{8}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \\
 &= (6\pi - 4\sqrt{2}) \text{ cm}^2
 \end{aligned}$$

e) $f'(x) = 2x + 7$

$$\begin{aligned}
 f(x) &= \int 2x + 7 \, dx \\
 &= x^2 + 7x + C
 \end{aligned}$$

sub (1, 4)

$$\begin{aligned}
 4 &= 1 + 7 + C \\
 C &= -4
 \end{aligned}$$

$$\therefore f(x) = x^2 + 7x - 4$$

f) $y = 3x^2 + 2x + 1$

$$y' = 6x + 2$$

at $x = 1$, $y' = 6 + 2 = 8$

$$\therefore y - 6 = 8(x - 1)$$

$$y - 6 = 8x - 8$$

$$y = 8x - 2$$

g) $e^{2x} + 3e^x - 10 = 0$

let $m = e^x$

$$m^2 + 3m - 10 = 0$$

$$(m+5)(m-2) = 0$$

$$m+5=0 \text{ or } m-2=0$$

$$m = -5 \quad m = 2$$

$$\therefore e^x = -5 \text{ or } e^x = 2$$

$$\begin{aligned}
 \text{no solns.} \quad \ln e^x &= \ln 2 \\
 x &= \ln 2
 \end{aligned}$$

Question 13

a) i) $m_{BC} = \frac{5-3}{1-0} = 2$

$$\begin{aligned}
 2x - y - 1 &= 0 \\
 y &= 2x - 1 \quad \therefore m_{AD} = 2
 \end{aligned}$$

$$\therefore BC \parallel AD$$

 $\therefore$ ABCD is a trapezium

(one pair of opposite sides parallel)

ii) sub $y = 5$ into $2x - y - 1 = 0$

$$2x - 5 - 1 = 0$$

$$2x = 6$$

$$x = 3$$

$$\therefore D(3, 5)$$

iii) $BC = \sqrt{(1-0)^2 + (5-3)^2}$

$$= \sqrt{1+4}$$

$$= \sqrt{5}$$

iv) $d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$

$$= \frac{|0 + -3 - 1|}{\sqrt{4+1}}$$

$$= \frac{4}{\sqrt{5}}$$

v) $d_{AD} = \sqrt{(3-0)^2 + (5+1)^2}$

$$= \sqrt{9+36}$$

$$= \sqrt{45}$$

$$= 3\sqrt{5}$$

$$A = \frac{4}{2\sqrt{5}} (3\sqrt{5} + \sqrt{5})$$

$$= 8 \text{ sq units.}$$

$$\begin{aligned}
 \text{b) } A &= \int_0^{\pi/2} 1 - \cos x \, dx + \int_{\pi/2}^{\pi} \sin x \, dx \\
 &= \left[x - \sin x \right]_0^{\pi/2} + \left[-\cos x \right]_{\pi/2}^{\pi} \\
 &= \left(\frac{\pi}{2} - \sin \frac{\pi}{2} \right) - (0 - \sin 0) \\
 &\quad + (-\cos \pi) - (-\cos \frac{\pi}{2})
 \end{aligned}$$

$$= \frac{\pi}{2} - 1 + 1$$

$$= \frac{\pi}{2} \text{ units}^2$$

c) $y = \ln \left[\frac{1-x}{1+x} \right]$

$$y = \ln(1-x) - \ln(1+x)$$

$$\frac{dy}{dx} = \frac{-1}{1-x} - \frac{1}{1+x}$$

$$= \frac{-(1+x) - (1-x)}{(1-x)(1+x)}$$

$$= \frac{-2}{1-x^2}$$

$$\int_0^{1/2} \frac{dx}{1-x^2}$$

$$= -\frac{1}{2} \int_0^{1/2} \frac{-2}{1-x^2} \, dx$$

$$= -\frac{1}{2} \left[\ln \left[\frac{1-x}{1+x} \right] \right]_0^{1/2}$$

$$= -\frac{1}{2} \left[\ln \frac{1/2}{3/2} - \ln \frac{1}{1} \right]$$

$$= -\frac{1}{2} \ln \frac{1}{3}$$

$$= -\frac{1}{2} (\ln 1 - \ln 3)$$

$$= \frac{\ln 3}{2}$$

Question 14

a) i) $\cos \theta \tan \theta = \cos \theta \frac{\sin \theta}{\cos \theta} = \sin \theta$

ii) $8 \sin \theta \cos \theta \tan \theta = \sec \theta$

$$8 \sin^2 \theta = \frac{1}{\sin \theta}$$

$$8 \sin^3 \theta = 1$$

$$\sin^3 \theta = \frac{1}{8}$$

$$\sin \theta = \frac{1}{2}, \theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

b) $f(x) = 3x^2 - x^3 + 9x - 2$

i) $f'(x) = 6x - 3x^2 + 9$

$f''(x) = 6 - 6x$

For stationary points, $f'(x) = 0$

$3x^2 - 6x - 9 = 0$

$x^2 - 2x - 3 = 0$

$(x-3)(x+1) = 0$

$x = 3, -1$

When $x = 3$,

$f(x) = 27 - 27 + 27 - 2 = 25$

and $f''(x) = 6 - 18 < 0$

$\therefore$ max at $(3, 25)$

When $x = -1$,

$f(x) = 3 + 1 - 9 - 2 = -7$

and $f''(x) = 6 + 6 > 0$

$\therefore$ min at $(-1, -7)$

ii) Possible point of inflexion

when $f''(x) = 0$

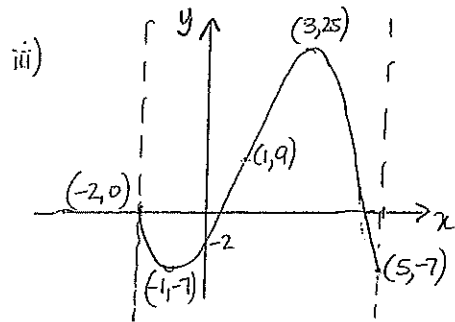
$6 - 6x = 0$

$x = 1$

test change in concavity

x	1^-	1	1^+
$f''(x)$	$+$	0	$-$

$\therefore$ point of inflexion at $(1, 9)$



iv) concave up for $-2 \leq x < 1$

c) $y = \ln(2x)$

$2x = e^y$

$x = \frac{e^y}{2}$

$V = \pi \int_1^6 x^2 dy$

$= \pi \int_1^6 \frac{e^{2y}}{4} dy$

$= \frac{\pi}{4} \left[\frac{e^{2y}}{2} \right]_1^6$

$= \frac{\pi}{8} [e^{12} - e^2] \text{ units}^3$

Question 15

a) $f(x) = \sqrt{25 - x^2}$

$25 - x^2 \geq 0$

$x^2 \leq 25$

$-5 \leq x \leq 5$ Domain

Range $0 \leq y \leq 5$

b) $x^2 - (k-2)x + 3(k-2) = 0$

no real roots $\rightarrow \Delta < 0$

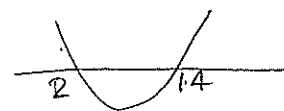
$b^2 - 4ac < 0$

$(k-2)^2 - 4 \cdot 3(k-2) < 0$

$k^2 - 4k + 4 - 12k + 24 < 0$

$k^2 - 16k + 28 < 0$

$(k-14)(k-2) < 0$



$\therefore 2 < k < 14$

c)

i) In ΔPQT and ΔRST

$\angle SET = \angle QPT$ (alternate angles
 $SR \parallel PQ$)

$\angle STE = \angle QTP$ (vertically opposite
angles equal)

$\therefore \frac{SR}{PQ} = \frac{RT}{PT}$ corresponding sides
in similar Δ s

and $\frac{RT}{PT} = \frac{VT}{UT}$ ratio of intercepts

$\therefore \frac{SR}{PQ} = \frac{VT}{UT}$

(ii) $\frac{SR}{4} = \frac{3-y}{y}$

$SR = \frac{12-4y}{y}$

$= \frac{12}{y} - 4$

iii) Area $\Delta PQR = \frac{1}{2} \times 4 \times y = 2y$

Area $\Delta RTS = \frac{1}{2} (3-y) \left(\frac{12}{y} - 4 \right)$
 $= \frac{1}{2} \left(\frac{36}{y} - 12 - 12 + 4y \right)$
 $= \frac{18}{y} - 12 + 2y$

$\therefore$ total area $= 4y - 12 + \frac{18}{y}$

iv) $A = 4y - 12 + 18y^{-1}$

$\frac{dA}{dy} = 4 - \frac{18}{y^2}$ $\frac{d^2A}{dy^2} = 36y^{-3}$

$\frac{dA}{dy} = 0$

$4 - \frac{18}{y^2} = 0$

$\frac{18}{y^2} = 4$

$y^2 = \frac{18}{4}$

$y = \pm \frac{\sqrt{3}}{2}$

disregard -ve as y is length

when $y = \frac{\sqrt{3}}{2}$, $\frac{d^2A}{dy^2} > 0$

$\therefore$ minimum occurs when $y = \frac{\sqrt{3}}{2}$

Question 16

a) i)

x	a	$2a$	$3a$
$\frac{1}{x}$	$\frac{1}{a}$	$\frac{1}{2a}$	$\frac{1}{3a}$

$$\int_a^{3a} \frac{1}{x} dx \div \frac{a}{3} \left[\frac{1}{a} + 4 \cdot \frac{1}{2a} + \frac{1}{3a} \right]$$

$$= \frac{a}{3} \left[-\frac{10}{3a} \right]$$

$$= \frac{10}{9}$$

ii) $\int_a^{3a} \frac{1}{x} dx = [\ln x]_a^{3a}$

$$= \ln 3a - \ln a$$

$$= \ln \frac{3a}{a}$$

$$= \ln 3. \quad \therefore \ln 3 \div \frac{10}{9}$$

b) $\ddot{x} = 4 \sin 2t$

i). $\dot{x} = \int 4 \sin 2t dt$

$$= -2 \cos 2t + C$$

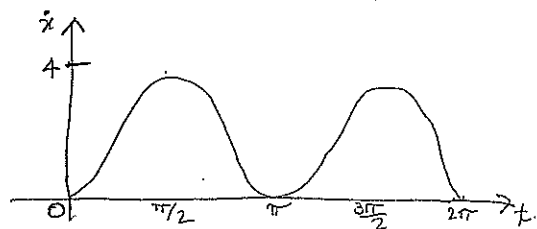
when $t=0$, $\dot{x}=0$

$$0 = -2 \cos 0 + C$$

$$C = 2$$

$$\therefore \dot{x} = 2 - 2 \cos 2t$$

ii) Period = $\frac{2\pi}{2} = \pi$ Range: $-1 \leq -\cos t \leq 1$
 $-2 \leq -2 \cos t \leq 2$
 $0 \leq 2 - 2 \cos t \leq 4$



at rest when $\dot{x}=0$
 i.e. $t=\pi$

iii).

$$d = \int_0^{\frac{2\pi}{3}} (2 - 2 \cos 2t) dt$$

$$= \left[2t - \sin 2t \right]_0^{\frac{2\pi}{3}}$$

$$= \left(\frac{4\pi}{3} - \sin \frac{4\pi}{3} \right) - (0 - \sin 0)$$

$$= \frac{4\pi}{3} + \frac{\sqrt{3}}{2}$$

c) i). $A = 15000 \times 1.04^7 + 5000$

ii) $A_2 = (15000 \times 1.04^7 + 5000) \times 1.04 + 5000$

$$= 15000 \times 1.04^8 + 5000 \times 1.04 + 5000$$

$$A_3 = (15000 \times 1.04^8 + 5000 \times 1.04 + 5000) \times 1.04 + 5000$$

$$= 15000 \times 1.04^9 + 5000 \times 1.04^2 + 5000 \times 1.04 + 5000$$

$$A_{10} = 15000 \times 1.04^{10} + 5000 \times 1.04^9 + 5000 \times 1.04^8 + \dots + 5000$$

$$= 15000 \times 1.04^{10} + 5000 (1 + 1.04 + 1.04^2 + \dots + 1.04^9)$$

$$= 15000 \times 1.04^{10} + 5000 \left[\frac{1(1.04^{10} - 1)}{1.04 - 1} \right]$$

$$= 28094.71869 + 5000 [12.00610712]$$

$$= 88125.2543$$

$$= \$88125.25$$